AEONTS

Non-Functional Requirement

for

Debt Operations System

Table of Contents

Document History

This section records the amendments made to this Questionnaire.

Version Number	Amended Date	Modified By	Description
1.0	25/10/2022	Nuttanun S.	Initial document
1.1	27/10/2022	Nuttanun S.	Update requirement

Non-Functional Information

This Non-Functional Requirement is aimed at capturing the initial and high-level requirements for preliminary estimates:

- Non-functional requirement
- Hardware requirement

Requirement Prioritization

MSCW	Definition	Description	Notes
M	Must have	Non-negotiable features that are mandatory for the business users and need to be completed at the earliest.	Highest
S	Should have	Important features that are not vital but add significant value.	Medium
C	Could have	Nice to have features that will have a small impact if left out.	Lower
W	Would have	Features that are not a priority for this specific time frame.	Lowest

Acronyms & Glossary of Terms

Term	Definition
VDI	Virtual Desktop Infrastructure
FTPS	File Transfer Protocol (over SSL)
SFTP	Secure File Transfer Protocol
AD	Active Directory
LDAP	Lightweight Directory Access Protocol
SOAP	Simple Object Access Protocol
SSO	Single Sign-on
XML	Extensible Mark-up Language
SIT	System Integration Test
UAT	User Acceptance Test

1 System Architecture / Server Configuration

1.1 Hardware Infrastructure Design

Item	Expectations	MoSCoW
1.1.1	To provide infrastructure design of the solution for AEONTS. Please include hardware specification of all the application servers, database servers, operating systems, certified platforms supported, etc.	M
1.1.2	Hardware Specification Environment • Data Center – Production Environment	M
1.1.3	• Data Center – User Acceptance Test Environment	M
1.1.4	• Data Center – System Integration Test Environment	M
1.1.5	• Data Center – Development Environment	M
1.1.6	• Disaster Recovery – Disaster Recovery Environment	M

1.2 High Availability

Item	Expectations	MoSCoW
1.2.1	‘Active-active’ availability same and across Production and Disaster Recovery is required. • As outlined in the following sub-sections, some components will be deployed and operated in an active-active configuration in production. All data will be backed up and recovered using standard AEON infrastructure services.	M
1.2.2	Supports automatic failover.	M
1.2.3	Capable to start/stop application partially.	M
1.2.4	Supports re-run Jobs (in case of errors).	M
1.2.5	As per AEON IT policy, 99.5% availability is expected, the Recovery Point Objective (RPO) is 30mins and the Recovery Time Objective (RTO) is 4hrs.	M
1.2.6	To enhance the availability in both Production and Disaster Recovery site, N+1 architecture is proposed so that even if one server is down/unavailable, still application can handle the expected load.	S

NOTE: Production Availability means that a system is on-line and accessible. A variety of factors can take a system off-line, ranging from planned downtime for maintenance to catastrophic failure. The goals of high availability solutions are to minimize this downtime and/or to minimize the time needed to recover from an outage. Exactly how much downtime can be tolerated will dictate the comprehensiveness, complexity and cost of the solution.

The table below shows the amount of downtime expected for different levels of availability:

Level of Availability	Downtime/Year
90%	40 days
95%	19 days
99%	4 days
99.9%	9 hours
99.95%	4 hours
99.99%	50 minutes
99.999%	5 minutes
99.9999%	30 seconds

CAUTION: To determine an appropriate level of availability we recommend first to define the high availability period required. For instance, system must be fully available from 8 a.m. to 6 p.m.

1.3 Software Requirement (Server)

Item	Expectations	MoSCoW
1.3.1	Software must be running 100% in virtual machine.	M
1.3.2	Software must be running between Data Center and Disaster Recovery site as active-active.	M
1.3.3	Software can be running between Data Center and Disaster Recovery site as active–passive for Database tier.	C
1.3.4	Software must be running on standard server hardening policies.	M
1.3.5	Software must be running on latest Operating System version.	M
1.3.6	Software must integrate with load balancing.	M

1.4 Software Requirement (Client)

Item	Expectations	MoSCoW
1.4.1	Application must be running on VDI system including citrix (both XenApp and Xendesk), VMWare, etc.	M
1.4.2	Application must be running on latest Operating System version.	M
1.4.3	Applicaion must be running on browser e.g. Google Chrome or Microsoft Edge latest version.	M
1.4.4	Screens Information for clients required the screen resolution is 1024x768.	W

2 Performance and scalability

2.1 Performance

Item	Expectations	MoSCoW
2.1.1	To provide tested for both transaction volume in normal state and in peak state.	W
2.1.2	To support multi-threaded operations (e.g. parallel batch processing) of all the batch jobs.	M
2.1.3	Able to support high volume transactions for peak periods i.e. end-of-year/quarter/month/ process. (Please refer to section 5. Data Volume)	M
2.1.4	To provide performance benchmarking in database performance category.	W
2.1.5	To provide performance benchmarking in business logic processing performance category.	W
2.1.6	To provide performance benchmarking in concurrent user performance category.	W

2.2 Scalability

Item	Expectations	MoSCoW
2.2.1	To proposed solution support scalability. (Horizontal/Vertical)	M
2.2.2	To provide solution support deployment on Cloud.	S
2.2.3	To provide commissioned independent scalability tests on a similar version/environment.	S

2.3 Data storage

Item	Expectations	MoSCoW
2.3.1	Capability to archive and access archived data. Please describe how the proposed solution provides access to archived data, and the archiving process or logic.	C
2.3.2	Capability to retain historical change records. The proposed system should be able to retain a historical change record of important business data. The contents of the change record should be retrieved easily.	C
2.3.3	Capability to store historical data. Please describe how historical data is stored, and whether there are any limitations (e.g. duration, volume, etc.) regarding the storage.	C
2.3.4	Capability to store Unicode characters. The proposed solution needs to be able to store Unicode that covers Thai languages.	M

3 Database

3.1 Database

Item	Expectations	MoSCoW
3.1.1	The system can be developed with Oracle Database on Linux and Red hat platform.	M
3.1.2	To propose database architecture to support the reliability and consistency of business data. Please describe how the database architecture of the proposed solution supports the reliability and consistency of the business data.	M
3.1.3	To provide database capacity planning.	M
3.1.4	Supports logs shipping to other sites e.g. Disaster Recovery or Replication.	M

3.2 Database scalability

Item	Expectations	MoSCoW
3.2.1	To support database scalability, data archiving and restoring database.	M
3.2.2	Capability for the database to be easily expanded.	M
3.2.3	To provide describe how the database architecture of the proposed solution supports business data volume growth, and how easily it can be expanded when data volume reaches a higher level.	M
3.2.4	Capability for database architecture to support large volume data loading. Please provide the technical details or how the proposed technology supports large data volume loading.	M

4 Integration

4.1 General

Item	Expectations	MoSCoW
4.1.1	Connectivity communication protocol can be supported on • HTTPS	M
4.1.2	• REST (HTTPS)	M
4.1.3	• SOAP (HTTPS)	M
4.1.4	• FTPS	M
4.1.5	• SFTP	M
4.1.6	Authentication method required to support Active Directory (AD) and Single sign-on (SSO).	M

4.2 Data integration

Item	Expectations	MoSCoW
4.2.1	Bulk data insert function with high performance when processing on high data volumes. (Please refer to section 5. Data Volume)	M
4.2.2	To supports batch integration.	M
4.2.3	To have Extract-Transform-Load (ETL) capability.	M
4.2.4	Transaction rollback functionality, regardless of the size or distribution of the transaction.	M
4.2.5	To supports Real-time integration.	M
4.2.6	Integrated task scheduling service.	M
4.2.7	Common set of audit, activity, and performance monitoring reports.	M
4.2.8	Integration specification must comply with existing interface at least 80%.	M

4.3 Batch Windows Time

Item	Expectations	MoSCoW
4.3.1	Monthly batch not exceed 8 hours.	M
4.3.2	Daily batch not exceed 4 hours.	M
4.3.3	Able to execute any batch anytime for example during working hour.	M

4.4 Response Time

Item	Expectations	MoSCoW
4.4.1	Support at least 1,000 transactions per second (calculate from batch process and online transactions process running parallel during peak period).	M
4.4.2	Maximum point to point response time is 0.5 second.	M
4.4.3	Maximum end to end response time is 2 seconds.	M
4.4.4	Timeout on web service calls is 30 seconds.	M

5 Data volume

5.1 Data Volume

Item	Expectations	MoSCoW
5.1.1	To support customer information at least 3 Millions.	M
5.1.2	To support account information at least 3.9 Millions.	M
5.1.3	To support active user at least 1,300 Users.	M

5.2 Transactions Volume

Item	Expectations	MoSCoW
5.2.1	To support loading payment transaction at least 150,000 transactions per hour.	M
5.2.2	To support integration with Telephony system at least 600,000 transactions per hour.	M
5.2.3	To support integration with Field Service system at least 10,000 transactions per hour.	M
5.2.4	To support integration with SMS system at least 100,000 transactions per hour.	M
5.2.5	To support integration with workflow system at least 2,000 transactions per hour.	M
5.2.6	To support login by integrating with active directory at least 1,300 transactions per hour.	M
5.2.7	To support loading pending payment transaction at least 60,000 transactions per hour.	M

5.3 Data Retention

Item	Expectations	MoSCoW
5.3.1	To collect activity history at least 7 months including current month in production.	M
5.3.2	To collect data retention at least 3 years.	M

6 Security

6.1 Security

Item	Expectations	MoSCoW
6.1.1	To provide information on what kind of security testing has been performed on the proposed solution.	C
6.1.2	Connectivity communication protocol can be supported on • HTTPS	M
6.1.3	• FTPS	M
6.1.4	• SFTP	M
6.1.5	• Encrypted password	M
6.1.6	• Fully encrypted transmission	M
6.1.7	To comply any related information security or IT security e.g. ISO27001, OWASP etc.	M

6.2 Data Encryption

Item	Expectations	MoSCoW
6.2.1	To ensure that all information complied with any related regulation for example PCIDSS, PDPA etc.	M

7 Authentication and authorization

7.1 Authentication

Item	Expectations	MoSCoW
7.1.1	Supports integration with Active Directory by LDAP.	M
7.1.2	Supports Single Sign-On.	M

7.2 Administration

Item	Expectations	MoSCoW
7.2.1	Granularity of access rights.	M
7.2.2	To provide feature assignment of simplified permission through a role hierarchy.	M

7.3 Login session control

Item	Expectations	MoSCoW
7.3.1	Systems must have session timeout management.	M
7.3.2	Systems must have idle management.	M

8 Migration

8.1 Data Migration

Item	Expectations	MoSCoW
8.1.1	To provide Migration strategy, Migration plan.	M
8.1.2	Perform data conversion / migration from existing systems to new system.	M

9 Backup, Restore and Disaster Recovery

9.1 Backup

Item	Expectations	MoSCoW
9.1.1	To recommend Daily, Weekly, Monthly, Yearly and ad hoc backups.	M
9.1.2	Ensure 100% replication of data between the Main Data Center and Disaster Recovery site.	M

9.2 Recovery

Item	Expectations	MoSCoW
9.2.1	Supports job re-run or sequential re-run.	M
9.2.2	Supports Bulk Data upload/download in short periods.	M
9.2.3	Separate specific environment components recovery.	M
9.2.4	Provide services available during windows maintenance or upgrades.	M
9.2.5	Application Recovery time 4 hour.	M

9.3 Disaster Recovery

Item	Expectations	MoSCoW
9.3.1	Supports disaster recovery on premise or cloud.	S
9.3.2	Supports Job re-run or sequential re-run.	M
9.3.3	Supports Bulk Data upload/download in short periods.	M
9.3.4	Separate specific environment components recovery.	M
9.3.5	Provide Services available during windows maintenance or upgrades.	M
9.3.6	Supports BCP (Business Continuity Plan).	M
9.3.7	Recovery sites (separate by system) • Hot site 30 minutes (active-active) • Warm site 4 hours	M

10 Batch process monitoring

10.1 Batch and background

Item	Expectations	MoSCoW
10.1.1	To provide batch processing lists with definitions, scheduling and parameters.	M
10.1.2	To provide batch auto-rerun functionality if the current attempt fails with parameter of times attempt.	M
10.1.3	To provide step by step manual runnable features.	M
10.1.4	To provide activate/deactivate by specified in advance start and expiry date and time.	C
10.1.5	To validation and define procedure in the solution for online processing and batch processing.	M
10.1.6	To support job re-run and job sequential step re-run.	M

11 Training and Deliverable

11.1 Design Deliverable

Item	Expectations	MoSCoW
11.1.1	After finish gathering requirement and design vendor must submit deliverable to ATS as follow : • Functional Design (Including Access matrix)	M
11.1.2	• Software Specification (Use case diagram/Sequence diagram/Activity diagram)	M
11.1.3	• System Design Specification (Entire System Solution/Component/Customized)	M
11.1.4	• Data Dictionary	M
11.1.5	• ER Diagram (Conceptual/Logical/Physical)	M
11.1.6	• Program Specification (System flow/Data flow/Development logic/SQL Statement)	M
11.1.7	• Integration Specification (Flat files, Web Services, API, etc)	M

11.2 Technical Deliverable

Item	Expectations	MoSCoW
11.2.1	After finish system architecture design vendor must submit deliverable to ATS as follow • Solution Architecture	M
11.2.2	• Capacity Plan	M
11.2.3	• Software Installation Manual (Preferably step by step documents)	M
11.2.4	• Network Architecture (Diagram / Schematics / Guide)	M
11.2.5	• Hardware Configuration (Diagram / Schematics / Guide)	M
11.2.6	• Database Configuration (Diagram / Schematics / Guide)	M

11.3 Implementation / Support Deliverable

Item	Expectations	MoSCoW
11.3.1	Before start system integration test vendor must submit deliverable to ATS as follow : • Configuration information, Source Code, Store Procedure, Indexing, Execution Script, etc.	M
11.3.2	• Test Plan, test script, test case and test result for SIT included functional and non-functional test.	M
11.3.3	• Test Plan, test script, test case for UAT included functional and non-functional test.	M
11.3.4	• Training for UAT	M

11.4 Release Software Deliverable (for each drop/release)

Item	Expectations	MoSCoW
11.4.1	Before deployment new release vendor must submit deliverable to ATS as follow : • Final checklist and readiness document.	M
11.4.2	• License summary list.	M
11.4.3	• Release Note.	M
11.4.4	• Release Control.	M
11.4.5	• System change control and impact analysis.	M
11.4.6	• Deployment Architecture.	M
11.4.7	• Deployment document (Step by step).	M
11.4.8	• System Limitation and Known issue/Workaround.	M

11.5 System Support Deliverable

Item	Expectations	MoSCoW
11.5.1	Before go-live vendor must submit deliverable to ATS as follow : • Administrator / Technical guide.	M
11.5.2	• User quick guide to troubleshoot common issues, Frequently Ask Questions (FAQ) e.g. batch processing / backup / adhoc script / knowledge base.	M
11.5.3	• Checklist of system maintenance.	M

11.6 Training (Thai Language)

Item	Expectations	MoSCoW
11.6.1	Before go-live vendor must prepare training material and training class for ATS as follow : • Training guide	M
11.6.2	• Training plan	M
11.6.3	• User manual – Operation (Supervisor, User)	M
11.6.4	• User manual - Management	M
11.6.5	• Technical Basic Administrator Training (On site – BKK) 1 class	M
11.6.6	• Backup/Recovery Training (On site – BKK) 1 class	M
11.6.7	• Performance tuning Training (On site – BKK) 1 class	M
11.6.8	• Operation Support Training (On site – BKK) 1 class	M
11.6.9	• Management User Training (On site – BKK) 2 classes	M
11.6.10	• Operation User Training (Train the trainer) (On site– BKK) 15 classes	M
11.6.11	• Support Operation user training (On site – 4 Centers) 50 classes	M

12 Monitoring

12.1 Monitoring

Item	Expectations	MoSCoW
12.1.1	To provide functions/tools for • Managing deadlocks in the system.	M
12.1.2	• Managing active and inactive user's sessions.	M
12.1.3	• Queuing tasks and determining schedules for business processes and administration tasks.	M
12.1.4	Able to integrate with other monitoring tools (Dynatrace, SolarWinds, PIM, etc).	M
12.1.5	Ability to provide automated monitoring result to specified user according to define criteria.	C

12.2 Application level audit trail

Item	Expectations	MoSCoW
12.2.1	To capture the information of data accessed (at file, row, column, or field levels).	M
12.2.2	To capture the information of before and after image of modified records.	M

12.3 User level audit trail

Item	Expectations	MoSCoW
12.3.1	To capture the information of User_ID, Transaction_ID & Transaction_Name, date and time, IP address, Workstation_ID, entries generated, document/file generated/assessed.	M

12.4 Audit trail change record

Item	Expectations	MoSCoW
12.4.1	To provide records of the changes in the audit trail include override user.	M
12.4.2	To provide the records of the changes in the audit trail include date and time of change.	M
12.4.3	To provide the records of the changes in the audit trail include value of the field before and after the change.	M

13 Support and Maintenance

13.1 Support

Item	Expectations	MoSCoW
13.1.1	Provide on-site system maintenance support (Thai Team) to support 24x7x365 at least 100 days for 3 years.	M
13.1.2	Provide on-site operation support (Thai Team) to support 8x7x365 for 1 year.	M
13.1.3	Provide Preventive Maintenance (PM) at least 4 times per year as routine basis, and ensure that all software/applications will run smoothly without downtime.	M

13.2 Service Level Agreement (SLA)

Item	Expectations	MoSCoW
13.2.1	Severity Critical: Response Time 15 minutes.	M
13.2.2	Severity Critical: Resolution Time 2 hours.	M
13.2.3	Severity High: Response Time 15 minutes.	M
13.2.4	Severity High: Resolution Time 4 hours.	M
13.2.5	Severity Medium: Response Time 1 hours.	M
13.2.6	Severity Medium: Resolution Time 8 hours.	M
13.2.7	Severity Low: Response Time 2 hours.	M
13.2.8	Severity Low: Resolution Time 3 days.	M

NOTE: Level of Severity

Severity	Description	Remark
Critical	Operation cannot process further and cannot use other workarounds. Business processes including data are heavily damage.	
High	Operation cannot process further, but can use other workarounds that needs supports from IT or Vendor.	All Critical severity must be completed before fixing High severity.
Medium	Operation can process, but leads to incorrect results.	All Critical and High severity must be completed before fixing Medium severity.
Low	No problems in operation, but contains usability or UI issues in the system Raise issue via ticket.	All Critical, High and Medium severity must be completed before fixing Medium severity.

13.3 Maintainability

Item	Expectations	MoSCoW
13.3.1	To provide error warning/pop-up message with configurable functions	M
13.3.2	To provide achieving and housekeeping of technical logs without any influence on the system integrity	M
13.3.3	To provide information to facilitate finding and fixing faults in the application.	M
13.3.4	To provide system maintenance reports i.e. utilization, transaction per period, DB, intuition detection.	M
13.3.5	To provide maintenance schedule.	M
13.3.6	To provide any standard support and maintenance processes (e.g. patch management, change management, upgrades, etc.) offered for the proposed solutions.	M

13.4 Solution upgrade

Item	Expectations	MoSCoW
13.4.1	To provide information and maintained and upgrade post implementation.	S

14 Other

14.1 Others

Item	Expectations	MoSCoW
14.1.1	To communicate changes in service offerings or roadmap and strategy.	S
14.1.2	To provide detail information about stress testing based on the proposed solution and recommended infrastructure.	C
14.1.3	Ability to ensure the AEONTS's data and services offered are not adversely affected if the vendor is involved in a merger, acquisition, or divestiture (Please explain)	M
14.1.4	Each party retains its rights in its pre-existing intellectual property. Except as set out in the applicable Statement of Work, any intellectual property developed for AEONTS in this project, and any working papers, source code compiled in connection with the services provided in this project, shall be the property of AEONTS	M